

Batch Normalization

A Short Tutorial for Deep Learning Students

Key Idea

Batch Normalization (BatchNorm) is a technique used in deep neural networks to normalize intermediate activations within a mini-batch. It generally makes training faster, more stable, and less sensitive to the choice of initialization.

1 Why Batch Normalization?

Consider a neural network layer that receives an input x and produces an intermediate value

$$z = Wx + b. \quad (1)$$

Here, W represents the weights and b represents the bias.

During training, the weights of previous layers continuously change. Consequently, the distribution and scale of the intermediate activations can also change. Batch Normalization helps control these activations by normalizing them using statistics computed from a mini-batch.

Intuition

Imagine that one layer produces values that are sometimes very small and sometimes very large. The next layer has to continuously adapt to these changes. BatchNorm keeps the activations in a more controlled form before passing them forward.

2 Input to Batch Normalization

Suppose a mini-batch contains m training examples and the layer produces d features. The activations can be represented by the matrix

$$Z = \begin{bmatrix} z_{11} & z_{12} & \cdots & z_{1d} \\ z_{21} & z_{22} & \cdots & z_{2d} \\ \vdots & \vdots & \ddots & \vdots \\ z_{m1} & z_{m2} & \cdots & z_{md} \end{bmatrix}. \quad (2)$$

The rows correspond to training examples and the columns correspond to features.

For each feature j , BatchNorm calculates its statistics across the m examples in the mini-batch.

3 Step 1: Compute the Batch Mean

For feature j , the mean is calculated as

$$\mu_j = \frac{1}{m} \sum_{i=1}^m z_{ij}. \quad (3)$$

Thus, every feature has its own mean.

For example, if the layer has three features, we calculate

$$\mu_1, \quad \mu_2, \quad \mu_3.$$

4 Step 2: Compute the Batch Variance

The variance for feature j is

$$\sigma_j^2 = \frac{1}{m} \sum_{i=1}^m (z_{ij} - \mu_j)^2. \quad (4)$$

The mean tells us where the activations are centered, while the variance tells us how much they are spread out.

Important

The mean and variance are computed separately for each feature, using the examples in the current mini-batch.

5 Step 3: Normalize the Activations

Once the mean and variance are available, each activation is normalized as

$$\hat{z}_{ij} = \frac{z_{ij} - \mu_j}{\sqrt{\sigma_j^2 + \epsilon}}, \quad (5)$$

where ϵ is a small positive number added for numerical stability.

The purpose of this operation is to approximately obtain

$$\mathbb{E}[\hat{z}_j] \approx 0, \quad \text{Var}(\hat{z}_j) \approx 1. \quad (6)$$

The normalized activations can be written as

$$\hat{Z} = \begin{bmatrix} \hat{z}_{11} & \hat{z}_{12} & \cdots & \hat{z}_{1d} \\ \hat{z}_{21} & \hat{z}_{22} & \cdots & \hat{z}_{2d} \\ \vdots & \vdots & \ddots & \vdots \\ \hat{z}_{m1} & \hat{z}_{m2} & \cdots & \hat{z}_{md} \end{bmatrix}. \quad (7)$$

6 Step 4: Scale and Shift

If BatchNorm always forced every feature to have mean 0 and variance 1, it could unnecessarily restrict what the network can learn.

Therefore, BatchNorm introduces two learnable parameters:

- γ_j : learnable scale parameter,
- β_j : learnable shift parameter.

The final BatchNorm output is

$$y_{ij} = \gamma_j \hat{z}_{ij} + \beta_j \quad (8)$$

The parameters γ_j and β_j are learned during training using backpropagation.

Intuition

Normalization provides a convenient standard form, while γ and β give the network the freedom to learn the scale and position that are useful for the task.

7 Complete Batch Normalization Formula

The complete process can be summarized as follows.

1. Compute the mean

$$\mu_j = \frac{1}{m} \sum_{i=1}^m z_{ij}. \quad (9)$$

2. Compute the variance

$$\sigma_j^2 = \frac{1}{m} \sum_{i=1}^m (z_{ij} - \mu_j)^2. \quad (10)$$

3. Normalize

$$\hat{z}_{ij} = \frac{z_{ij} - \mu_j}{\sqrt{\sigma_j^2 + \epsilon}}. \quad (11)$$

4. Scale and shift

$$y_{ij} = \gamma_j \hat{z}_{ij} + \beta_j \quad (12)$$

Thus,

$$z \longrightarrow \text{Mean/Variance} \longrightarrow \text{Normalize} \longrightarrow \text{Scale and Shift} \longrightarrow y \quad (13)$$

8 BatchNorm in a Neural Network

A common arrangement in a neural network is

$$\boxed{\text{Input} \rightarrow \text{Linear Layer} \rightarrow \text{BatchNorm} \rightarrow \text{ReLU} \rightarrow \text{Next Layer}} \quad (14)$$

For example,

$$x \rightarrow (Wx + b) \rightarrow \text{BatchNorm} \rightarrow \text{ReLU}.$$

The BatchNorm operation is therefore performed on the intermediate activations before they are passed to the activation function and the subsequent layer.

9 Training Versus Inference

One of the most important concepts in BatchNorm is that the statistics used during training and inference are different.

9.1 During Training

During training, the mean and variance are calculated from the current mini-batch:

$$\mu_j = \mu_{j,\text{batch}}, \quad \sigma_j^2 = \sigma_{j,\text{batch}}^2. \quad (15)$$

At the same time, BatchNorm maintains running estimates of these statistics, commonly using an exponential moving average.

9.2 During Inference

During testing or inference, the model should not depend on the particular examples present in a mini-batch. Therefore, the stored running estimates are used:

$$y = \gamma \frac{z - \mu_{\text{running}}}{\sqrt{\sigma_{\text{running}}^2 + \epsilon}} + \beta. \quad (16)$$

The learned γ and β parameters are used as fixed parameters during inference.

Important

A common mistake is to use the statistics of the current test example during inference. BatchNorm normally uses the running mean and variance estimated during training.

10 Benefits of Batch Normalization

BatchNorm can provide several practical benefits:

1. **Faster convergence:** Training often converges more quickly.

2. **More stable training:** The scale of intermediate activations is controlled.
3. **Higher learning rates:** The network can often tolerate relatively larger learning rates.
4. **Regularization effect:** The randomness introduced by mini-batch statistics can sometimes reduce overfitting.
5. **Reduced sensitivity to initialization:** Training can become less sensitive to the initial values of weights.

11 What Does BatchNorm Actually Normalize?

This is an important conceptual distinction.

Quick Summary

BatchNorm normalizes activations, not weights.

For a layer

$$z = Wx + b,$$

BatchNorm operates on z (or the corresponding layer activations), rather than directly normalizing W .

This is different from techniques such as weight decay or L_2 regularization, which directly affect the magnitude of the weights.

12 Simple Numerical Example

Consider one feature whose activations in a mini-batch are

$$z = [2, 4, 6].$$

The batch mean is

$$\mu = \frac{2 + 4 + 6}{3} = 4. \tag{17}$$

The variance is

$$\sigma^2 = \frac{(2 - 4)^2 + (4 - 4)^2 + (6 - 4)^2}{3} = \frac{8}{3}. \tag{18}$$

Ignoring the very small ϵ for simplicity, the normalized values are

$$\hat{z} = \left[\frac{-2}{\sqrt{8/3}}, \frac{0}{\sqrt{8/3}}, \frac{2}{\sqrt{8/3}} \right]. \tag{19}$$

Therefore, the values are centered around zero and have approximately unit variance.

If the learned parameters are, for example,

$$\gamma = 2, \quad \beta = 1,$$

then

$$y = 2\hat{z} + 1. \tag{20}$$

Thus, normalization is followed by a learned scale and shift.

13 BatchNorm: Four Steps to Remember

Students can remember BatchNorm using the following four-step sequence:

1. **Mean:** calculate the batch mean.
2. **Variance:** calculate the batch variance.
3. **Normalize:** obtain approximately zero mean and unit variance.
4. **Scale + Shift:** use learnable γ and β .

In compact form,

$$\boxed{\text{Mean} \rightarrow \text{Variance} \rightarrow \text{Normalize} \rightarrow \text{Scale} + \text{Shift}} \tag{21}$$

14 Key Points for Examination

- BatchNorm normalizes intermediate activations.
- Mean and variance are calculated for each feature across a mini-batch during training.
- ϵ is added for numerical stability.
- γ and β are learnable parameters.
- γ controls scale and β controls shift.
- During inference, running estimates of mean and variance are used.
- BatchNorm can improve training stability and convergence.
- BatchNorm is not the same as weight regularization.

15 One-Line Definition

Batch Normalization normalizes the intermediate activations of a neural network using mini-batch statistics and then restores useful scale and shift through learnable parameters γ and β .